

PENCH FOODS - LOGISTICS ARCHITECTURE

State Machine, Trip Lifecycle, and API Integration Blueprint

THE THREE LIFECYCLE PILLARS

☐ **is_locked (Boolean)**

Defines route stability. Controls whether stop configurations and order quantities can be edited.

- false: Open for changes (Client Price overrides, sequence adjustments, item swaps).
- true: Frozen for loading (Trips cannot be modified by admins or customers). Locked at 6:00 AM.

☐ **status (TextChoices Choice Field)**

Represents the active physical dispatch stage of the delivery driver on the road.

- 'pending': Created/Optimized. Rider has not departed the hub.
- 'started' / 'in_progress': Rider is out for delivery. GPS coordinates tracking is active.
- 'completed': Rider successfully ended trip. • 'stopped': Aborted manually or auto-terminated.

☐ ☐ **is_completed (Boolean)**

A database query optimizer flag to quickly separate active trips from archived history.

- false: Active, pending, or in-progress trip. Loaded inside the driver's active panel.
- true: Historically closed. Removed from driver console; frees driver for new assignment.

BUSINESS RULES & TIME GATEKEEPERS

☐ **12:00 AM (Midnight) - Automatic Generation Task**

System loops through all active customer subscriptions and generates tomorrow's orders. Uses geographic TSP Or-Tools solver to map orders into optimized sequence routes.

☐ **06:00 AM (Morning) - Pre-Delivery Lock Cutoff**

System auto-freezes all pending routes for the day (is_locked -> true). Primes stock totals and stops loading, preventing any post-departure modifications.

☐ **12:00 PM (Noon) - Auto-Stop Termination Cutoff**

Auto-terminates active trips that failed to complete (status -> 'stopped', is_completed -> true). Stops remaining are marked 'undelivered', releasing riders back into the pool.

TRIP STATE MACHINE - DATA MATRIX

Detailed State Transitions, Values, and Frontend Mapping Matrix

TRIP TRANSITIONS MATRIX

Operational Stage	status	is_locked	is_completed	API Endpoint
1. Route Generated	"pending"	false	false	/generate/
2. Locked (06:00 AM)	"pending"	true	false	/lock/
3. Started Trip	"started"	true	false	/start-trip/
4. Completed Trip	"completed"	true	true	/stop-trip/
5. Aborted (12:00 PM)	"stopped"	true	true	Celery beat auto

DEVELOPMENT GUIDELINES & BEST PRACTICES

UI Badging Guidelines (Admin Web & Rider Mobile)

- If is_locked is true -> Show standard Lock Icon next to the Route Card Name header.
- If status is 'started' or 'in_progress' -> Show glowing pulsating Green Dot (Rider active on road).
- If status is 'completed' -> Show checked Emerald badge (Archived successfully).

Web Admin - Action Control Interlocks

- Only show 'Lock Route' button on Web Panel if is_locked is false.
- Re-enable 'Unlock Route' button so Admin can override/unlock a locked route manually.
- Disable ALL order/quantities/stops editing forms in the UI instantly if is_locked is true.

Live Driver Tracking & WebSockets

- Path: ws://<tenant-subdomain>.<clean-host>/ws/tracking/
- Mobile app starts WebSocket socket only when driver clicks 'Start Trip' and stops on completion.
- Send flat geolocation updates: { 'latitude': 21.145, 'longitude': 79.088, 'heading': 180.0 }.

Dedicated Mobile API

- Endpoints: GET /api/drivers/my-route/ and GET /api/drivers/trip-status/
- Backend returns the oldest active, incomplete route assigned to the authenticated driver user.
- High compatibility: BOTH 'id' and 'route_id' are returned to avoid mobile parser faults.